

Numerical Methods for Partial Differential Equations

Alfio Quarteroni

MOX, Dipartimento di Matematica
Politecnico di Milano



Lecture Notes
A.Y. 2025-2026

Boundary value problems

Consider

$$\begin{cases} \mathcal{L}u = f & \text{in } \Omega \\ + \text{ B.C.} & \text{on } \partial\Omega \end{cases} \quad (1)$$

- Ω : open bounded domain in $\mathbb{R}^d$, with $d = 2, 3$
- $\partial\Omega$: boundary of Ω
- f : given
- B.C. to be prescribed according to $\mathcal{L}$
- $\mathcal{L}$: 2nd order differential operator. Examples:

1 $\mathcal{L}u = -\operatorname{div}(\mu \nabla u) + \mathbf{b} \cdot \nabla u + \sigma u$ (non-conservative form)

2 $\mathcal{L}u = -\operatorname{div}(\mu \nabla u) + \operatorname{div}(\mathbf{b}u) + \sigma u$ (conservative form)

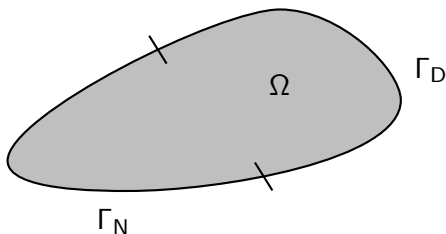
$$\mu \in L^\infty(\Omega), \mu(\mathbf{x}) \geq \mu_0 > 0 \qquad \sigma \in L^2(\Omega)$$

$$\mathbf{b} \in (L^\infty(\Omega))^d \qquad f \in L^2(\Omega)$$

Example

$$\begin{cases} \mathcal{L}u = -\operatorname{div}(\mu \nabla u) + \mathbf{b} \cdot \nabla u + \sigma u = f & \text{in } \Omega \\ u = 0 & \text{on } \Gamma_D \\ \mu \nabla u \cdot \mathbf{n} = g & \text{on } \Gamma_N \end{cases} \quad (2)$$

$$g \in L^2(\Gamma_N), \quad \partial\Omega = \Gamma_D \cup \Gamma_N, \quad \dot{\Gamma}_D \cap \dot{\Gamma}_N = \emptyset$$



Weak formulation

Weak form of Eq. (2)

Idea: v = suitable test function $\rightarrow$ multiply (2) by v : $(\mathcal{L}u) v = f v$
 $\rightarrow$ integrate in $\Omega \rightarrow$ apply integration by parts

$$\underbrace{\int_{\Omega} \mu \nabla u \cdot \nabla v + \int_{\Omega} \mathbf{b} \cdot \nabla u v + \int_{\Omega} \sigma u v}_{=: a(u, v)} = \int_{\Omega} f v + \underbrace{\int_{\Gamma_D} \mu \nabla u \cdot \mathbf{n} v}_{=0 \text{ if } v|_{\Gamma_D} = 0} + \int_{\Gamma_N} \underbrace{\mu \nabla u \cdot \mathbf{n} v}_{=g} \quad (3)$$

whence: $\left\{ \begin{array}{l} \text{Find } u \in V = \{v \in H^1(\Omega), v|_{\Gamma_D} = 0\} =: H_{\Gamma_D}^1(\Omega) \\ a(u, v) = \langle F, v \rangle \quad \forall v \in V \end{array} \right. \quad (4)$

■ $a: V \times V \rightarrow \mathbb{R}$ bilinear form

■ $F: V \rightarrow \mathbb{R}$ linear form s.t. $\langle F, v \rangle \equiv F(v) = \int_{\Omega} f v + \int_{\Gamma_N} g v$

Lax-Milgram Lemma

Sufficient conditions for existence and uniqueness of the solution

Theorem (Lax-Milgram)

Assume that:

- i V Hilbert space with norm $\|\cdot\|$ and inner product $(\cdot, \cdot)$
- ii $F \in V'$: $|F(v)| \leq \|F\|_{V'} \|v\| \quad \forall v \in V$
- iii a continuous: $\exists M > 0$: $|a(u, v)| \leq M \|u\| \|v\| \quad \forall u, v \in V$
- iv a coercive: $\exists \alpha > 0$: $a(v, v) \geq \alpha \|v\|^2 \quad \forall v \in V$

Then, there exists a unique solution u of (4).

Remark: $V' =$ dual space of $V = \{ \text{linear and bounded (i.e. continuous), maps from } V \text{ to } \mathbb{R} \}$ with norm:

$$\|F\|_{V'} = \sup_{v \in V \setminus \{0\}} \frac{|F(v)|}{\|v\|_V}$$

Moreover:

$$\alpha \|u\|^2 \leq a(u, u) = F(u) \leq \|F\|_{V'} \|u\|$$

Hence:

$$\|u\| \leq \frac{\|F\|_{V'}}{\alpha} \rightarrow \text{stability / continuous dependence on data}$$

Questions:

- How to prove assumptions of Lax-Milgram Lemma?
→ case by case
- what if the assumptions are not satisfied (in particular, iv)?
→ try Nečas Theorem

Nečas theorem

To formulate Nečas Theorem, consider a slightly more general problem than (4):

Problem

$$\begin{cases} \text{Find } u \in V \\ a(u, w) = \langle F, w \rangle \quad \forall w \in W \end{cases} \quad (5)$$

Note: $\langle F, w \rangle$ is a different (equivalent) notation for $F(w)$.

$a: V \times W \rightarrow \mathbb{R}$ bilinear form

$F: W \rightarrow \mathbb{R}$ linear and continuous (that is $F \in W'$)

Theorem (Nečas)

Assume that $F \in W'$. Consider the following conditions.

A *a continuous*: $\exists M > 0: |a(u, w)| \leq M \|u\|_V \|w\|_W \forall u \in V, w \in W$

B *inf-sup condition*: $\exists \alpha > 0: \forall v \in V \sup_{w \in W \setminus \{0\}} \frac{a(v, w)}{\|w\|_W} \geq \alpha \|v\|_V$

C $\forall w \in W, w \neq 0, \exists v \in V: a(v, w) \neq 0$

Conditions **A**–**B**–**C** are necessary and sufficient for the existence and uniqueness of a solution of (5), for any $F \in W'$.

Moreover (continuous dependence on data):

$$\|u\|_V \leq \frac{1}{\alpha} \|F\|_{W'} \quad (6)$$

Proof of Thm. 3: see [S].

N.B. **B** can be restated as: $\exists \alpha > 0: \inf_{v \in V} \sup_{w \in W \setminus \{0\}} a(v, w) \geq \alpha \|v\|_V \|w\|_W$

[S] S. Salsa, *Partial Differential Equations in Action – from Modelling to Theory*, Springer, 2008

Remark

If $W = V$, then Thm. 3 provides necessary and sufficient conditions for existence and uniqueness of solutions of problem. 4 (as a special case).

Note that in this case:

A $\iff$ **iii**

Whereas:

B yields:

$$\exists \alpha > 0: \forall v \in V \quad \sup_{w \in V \setminus \{0\}} \frac{a(v, w)}{\|w\|_V} \geq \alpha \|v\|_V \quad (7)$$

C yields:

$$\forall w \in V, w \neq 0, \exists v \in V: a(v, w) \neq 0 \quad (8)$$

(7) and (8) are more general / weaker conditions than coercivity (i.e. **iv**).
Indeed, by taking $w = v$, **iv** $\implies$ (7), (8)

Galerkin method (for problem (4))

$$\text{Find } u_h \in V_h: a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h \quad (\text{G})$$

Petrov-Galerkin method (for problem (5))

$$\text{Find } u_h \in V_h: a(u_h, w_h) = F(w_h) \quad \forall w_h \in W_h \quad (\text{PG})$$

where:

- $\{V_h, h > 0\}$ finite dimensional subspaces of V
- $\{W_h, h > 0\}$ finite dimensional subspaces of W
- $\dim V_h = \dim W_h = N_h < +\infty$

Analysis of (G)

Existence and uniqueness

Corollary of Lax-Milgram Lemma (V_h being a closed subspace of V)

Stability

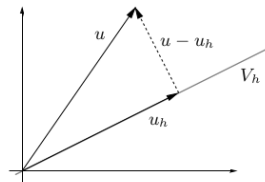
Uniform bound w.r.t. h :

$$\|u_h\| \leq \frac{\|F\|_{V'}}{\alpha}$$

Consistency = Galerkin orthogonality

By subtracting $a(u_h, v_h) = F(v_h)$ from $a(u, v_h) = F(v_h)$ we obtain:

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h$$



Convergence (Céa Lemma)

$$\begin{aligned}\alpha \|u - u_h\|^2 &\leq a(u - u_h, u - u_h) \\ &= a(u - u_h, u - v_h) + \underbrace{a(u - u_h, v_h - u_h)}_{=0 \text{ (Galerkin orthogonality)}} \\ &\leq M \|u - u_h\| \|u - v_h\| \quad \forall v_h \in V_h\end{aligned}$$

Hence:

$$\|u - u_h\| \leq \frac{M}{\alpha} \|u - v_h\| \quad \forall v_h \in V_h$$

Finally:

$$\|u - u_h\| \leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|$$

Analysis of (G)

Assumption: *space saturation*

$$\forall v \in V \quad \lim_{h \rightarrow 0} \inf_{v_h \in V_h} \|v - v_h\| = 0$$

Then: $\lim_{h \rightarrow 0} \|u - u_h\| = 0$ convergence

Rate of convergence (ex: Finite Elements)

$\mathcal{T}_h = \bigcup K$: triangulation of Ω (of Ω_h actually)

$$V_h = \{v_h \in C^0(\overline{\Omega}) : v_h|_K \in \mathbb{P}^r(K) \forall K \in \mathcal{T}_h, v_h|_{\Gamma_D} = 0\}, \quad \text{for } r \geq 1$$

$$\inf_{v_h \in V_h} \|u - v_h\| \leq \|u - \bar{u}_h\| \quad (\text{suitable choice of } \bar{u}_h)$$

Example: $\bar{u}_h = \Pi_h^r u$ Finite Element interpolant

$$\|u - \bar{u}_h\| \leq C h^r |u|_{H^{r+1}(\Omega)}$$

provided $u \in V \cap H^{r+1}(\Omega)$.

Examples of polynomial spaces

$$\text{1D } (\mathbb{P}^r): \quad p(x) = \sum_{k=0}^r a_k x^k$$

$$\text{2D } (\mathbb{P}^r): \quad p(x_1, x_2) = \sum_{\substack{k=0,\dots,r \\ m=0,\dots,r \\ k+m \leq r}} a_{km} x_1^k x_2^m$$

$$\text{3D tetrahedra } (\mathbb{P}^r): \quad p(x_1, x_2, x_3) = \sum_{\substack{k=0,\dots,r \\ m=0,\dots,r \\ n=0,\dots,r \\ k+m+n \leq r}} a_{kmn} x_1^k x_2^m x_3^n$$

$$\text{3D hexahedra } (\mathbb{Q}^r): \quad p(x_1, x_2, x_3) = \sum_{\substack{k=0,\dots,r \\ m=0,\dots,r \\ n=0,\dots,r}} a_{kmn} x_1^k x_2^m x_3^n$$

Basis functions

Having defined a basis $\{\phi_j(\mathbf{x})\}_{j=1}^{N_h}$ for the space V_h , each function $v_h \in V_h$ can be expanded as a linear combination of elements of the basis, suitably weighted by the coefficients $\{v_j\}_{j=1}^{N_h}$:

$$v_h(\mathbf{x}) = \sum_{j=1}^{N_h} v_j \phi_j(\mathbf{x})$$

We will use the notation $\mathbf{v} = (v_1, \dots, v_{N_h})^T$ to denote a vector $\mathbf{v} \in \mathbb{R}^{N_h}$ collecting all the basis coefficients (also called *degrees of freedom*).

A basis is called **lagrangian** if it satisfies the following property:

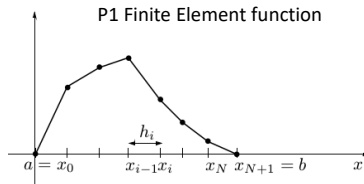
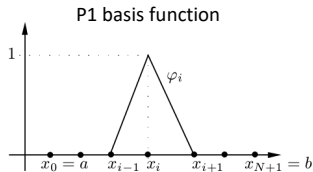
$$\phi_i(\mathbf{x}_j) = \delta_{ij} \quad \forall 1 \leq i, j \leq N_h$$

for a suitable collection of points $\{\mathbf{x}_j\}_{j=1}^{N_h}$ called *nodes*. When the basis is lagrangian, the following property holds:

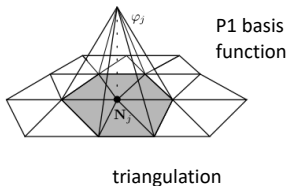
$$v_h(\mathbf{x}_j) = v_j \quad \forall 1 \leq j \leq N_h$$

Examples of triangulations and FE functions

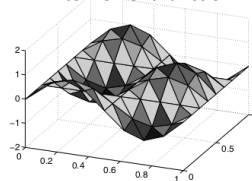
1D



2D



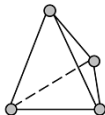
P1 Finite Element function



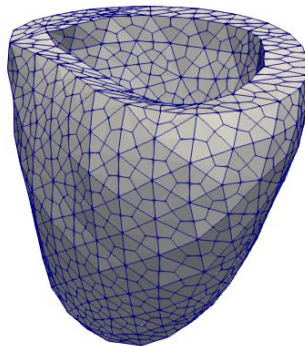
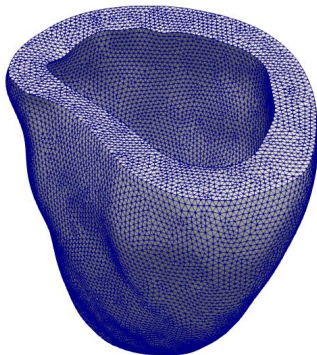
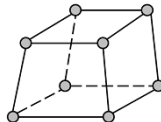
Examples of triangulations and FE functions

3D

tetrahedra



hexahedra



Example (back to Problem 2)

Proof of conditions of Thm. 2 (Lax-Milgram)

Condition i

V Hilbert space, as it is a closed subspace of the Hilbert space $H^1(\Omega)$.

$(u, v) = \int_{\Omega} u v + \int_{\Omega} \nabla u \cdot \nabla v$ scalar product

$\|v\| = (\int_{\Omega} v^2 + \int_{\Omega} |\nabla v|^2)^{1/2}$ norm

Condition iii ($F \in V'$)

$$\begin{aligned} |\langle F, v \rangle| &= \left| \int_{\Omega} f v + \int_{\Gamma_N} g v \right| \\ &\leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\Gamma_N)} \|v\|_{L^2(\Gamma_N)} \quad (\text{Cauchy-Schwarz inequality}) \\ &\leq \|f\|_{L^2(\Omega)} \|v\| + \|g\|_{L^2(\Gamma_N)} C_{\text{trace}} \|v\| \quad (\text{definition of } \|\cdot\| \text{ and trace ineq.}) \\ &= (\|f\|_{L^2(\Omega)} + C_{\text{trace}} \|g\|_{L^2(\Gamma_N)}) \|v\| \end{aligned}$$

$$\|F\|_{V'} = \sup_{v \neq 0} \frac{|\langle F, v \rangle|}{\|v\|} \leq \|f\|_{L^2(\Omega)} + C_{\text{trace}} \|g\|_{L^2(\Gamma_N)} < +\infty$$

Example (back to Problem 2)

Proof of conditions of Thm. 2 (Lax-Milgram)

Condition iii (a continuous)

$$\begin{aligned} |a(u, v)| &\leq \|\mu\|_{L^\infty} \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} + \|\mathbf{b}\|_{L^\infty} \|\nabla u\|_{L^2} \|v\|_{L^2} + \|\sigma\|_{L^2} \|u\|_{L^4} \|v\|_{L^4} \\ &\leq \underbrace{(\|\mu\|_{L^\infty} + \|\mathbf{b}\|_{L^\infty} + \|\sigma\|_{L^2})}_M \|u\| \|v\| \quad \forall u, v \in V \end{aligned}$$

N.B. Sobolev embedding: $\|v\|_{L^4} \leq \|v\| \quad \forall v \in H^1$

Example (back to Problem 2)

Proof of conditions of Thm. 2 (Lax-Milgram)

Condition iv (a coercive)

This condition holds under the assumptions:

$$\sigma - \frac{1}{2} \operatorname{div} \mathbf{b} \geq 0 \text{ in } \Omega, \quad \mathbf{b} \cdot \mathbf{n} \geq 0 \text{ on } \Gamma_N \quad (9)$$

$$\begin{aligned} a(v, v) &= \int_{\Omega} \mu |\nabla v|^2 + \underbrace{\int_{\Omega} \mathbf{b} \cdot \frac{1}{2} \nabla (v^2)}_{\int_{\Omega} -\frac{1}{2} \operatorname{div} \mathbf{b} (v^2) + \frac{1}{2} \int_{\partial\Omega} \mathbf{b} \cdot \mathbf{n} v^2} + \int_{\Omega} \sigma v^2 \\ &= \int_{\Omega} \mu |\nabla v|^2 + \int_{\Omega} \left(\sigma - \frac{1}{2} \operatorname{div} \mathbf{b} \right) v^2 + \frac{1}{2} \int_{\Gamma_N} \mathbf{b} \cdot \mathbf{n} v^2 \\ &\geq \mu_0 \|\nabla v\|_{L^2}^2 \end{aligned}$$

Note: if $\mathbf{b}$ is constant, then $\operatorname{div} \mathbf{b} = 0$, and the first equation in (9) reduces to $\sigma \geq 0$.

Example (back to Problem 2)

Proof of conditions of Thm. 2 (Lax-Milgram)

Poincaré inequality

If Γ_D is a set of positive measure (in 1D, it is sufficient that Γ_D contains a single point), then:

$$\exists C_P > 0: \|v\|_{L^2(\Omega)} \leq C_P \|\nabla v\|_{L^2(\Omega)} \quad \forall v \in H_{\Gamma_D}^1(\Omega)$$

Then: $\|v\|^2 = \|v\|_{L^2(\Omega)}^2 + \|\nabla v\|_{L^2(\Omega)}^2 \leq (1 + C_P^2) \|\nabla v\|_{L^2(\Omega)}^2$

Whence: $\|\nabla v\|_{L^2(\Omega)}^2 \geq (1 + C_P^2)^{-1} \|v\|^2$

In conclusion (coercivity):

$$a(v, v) \geq \frac{\mu_0}{1 + C_P^2} \|v\|^2$$

Remark (not for exam): what if (9) is not satisfied?

Try with Nečas Theorem.

The following assumptions are equivalent to those of Nečas Theorem:

a Weak coercivity (Gårding inequality)

$$\exists \alpha, \lambda > 0: a(v, v) \geq \alpha \|v\|^2 - \lambda \|v\|_{L^2(\Omega)}^2 \quad \forall v \in V$$

b Uniqueness condition:

$$(a(u, v) = 0 \forall v \in V) \implies u = 0$$

Typically proven by maximum principle

Similar result in finite dimension. Consider problem (G).

Condition **a** holds in the subspace as well:

$$\exists \alpha, \lambda > 0: a(v_h, v_h) \geq \alpha \|v_h\|^2 - \lambda \|v_h\|_{L^2(\Omega)}^2 \quad \forall v_h \in V_h$$

Condition **a** does not imply:

$$(a(u_h, v_h) = 0 \forall v_h \in V_h) \implies u_h = 0.$$

the latter property needs to be proven!

Conditioning of the stiffness matrix

Reminder: if A is symmetric positive definite (spd), then

$$K_2(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$$

Proposition

If $a(\cdot, \cdot)$ is symmetric and coercive, then A is spd

Proof.

Symmetry: $A_{ij} = a(\phi_j, \phi_i) = a(\phi_i, \phi_j) = A_{ji}$

For any $\mathbf{v} \in \mathbb{R}^{N_h}$:

$$\begin{aligned}\mathbf{v}^T A \mathbf{v} &= \sum_{i,j} A_{ij} v_i v_j = \sum_{i,j} a(\phi_j, \phi_i) v_i v_j \\ &= a\left(\sum_j v_j \phi_j, \sum_i v_i \phi_i\right) = a(v_h, v_h) \geq \alpha \|v_h\|^2 > 0\end{aligned}$$

if $(v_h \neq 0 \iff \mathbf{v} \neq \mathbf{0})$. Hence, A is pos. def. □

Definition

If A is spd, we define the A -norm of $\mathbf{v}$ as

$$\|\mathbf{v}\|_A := (A\mathbf{v}, \mathbf{v})^{1/2} = \left(\sum_{i,j} a_{ij} v_i v_j \right)^{1/2}$$

Conditioning of the stiffness matrix

We can prove that $\exists C_1, C_2 > 0: \forall \lambda_h$ eigenvalue of A :

$$\alpha C_1 h^d \leq \lambda_h \leq M C_2 h^{d-2} \quad d = 1, 2, 3$$

whence:

$$\frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \leq \frac{M C_2}{\alpha C_1} h^{-2}$$

This is indeed an asymptotic estimate (also a lower bound, with a different constant). Then:

$$K_2(A) = \mathcal{O}(h^{-2})$$

Implication. If we use the conjugate gradient method to solve $A\mathbf{u} = \mathbf{f}$, then:

$$\|\mathbf{u}^{(k)} - \mathbf{u}\|_A \leq 2 \left(\frac{\sqrt{K_2(A)} - 1}{\sqrt{K_2(A)} + 1} \right)^k \|\mathbf{u}^{(0)} - \mathbf{u}\|_A$$

Same with gradient method, with $\sqrt{K_2(A)}$ replaced by $K_2(A)$.
 $\implies$ need for preconditioners!

Remark. How many iterations to reduce the relative error by a factor 10 with the gradient method?

$$\left(\frac{K_2(A) - 1}{K_2(A) + 1} \right)^k < 10^{-1} \iff k > \frac{\log(10)}{\log \left(\frac{K_2(A) - 1}{K_2(A) + 1} \right)}$$

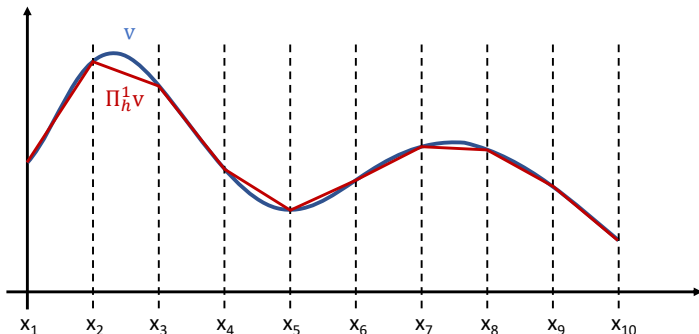
With the conjugate gradient method:

$$k > \frac{\log(10)}{\log \left(\frac{\sqrt{K_2(A)} - 1}{\sqrt{K_2(A)} + 1} \right)}$$

Interpolation Error Estimate

Use continuous FE of degree $r \geq 1$. Assume $v \in H^{r+1}(\Omega)$ let $\Pi_h^r v$ be the finite element interpolant of v at the FE nodes, that is:

$$\Pi_h^r v \in X_h^r, \quad \Pi_h^r(\mathbf{x}_i) = v(\mathbf{x}_i) \quad \forall \text{ node } \mathbf{x}_i \text{ of } \mathcal{T}_h$$



Interpolation Error Estimate

Then for $m = 0, 1$, $\exists C = C(r, m, \hat{k})$ such that:

$$|v - \Pi_h^r v|_{H^m(\Omega)} \leq C \left(\sum_{K \in \mathcal{T}_h} h_K^{2(r+1-m)} |v|_{H^{r+1}(K)}^2 \right)^{1/2} \quad (10)$$

$$h_K = \text{diam}(K)$$

since $h_K \leq h \quad \forall K$, this yields:

$$|v - \Pi_h^r v|_{H^m(\Omega)} \leq C h^{r+1-m} |v|_{H^{r+1}(K)} \quad \forall v \in H^{r+1}(\Omega), m = 0, 1 \quad (11)$$

Note: $H^0(\Omega) = L^2(\Omega)$, so that $|v|_{H^0(\Omega)} = \|v\|_{L^2(\Omega)} = \sqrt{\int_{\Omega} |v|^2 dx}$.

Finite Element error estimate

Recall that:

$$\begin{aligned}\|u - u_h\| &= \|u - u_h\|_{H^1(\Omega)} \\ &\leq \frac{M}{\alpha} \inf_{v_h \in V_h} \|u - v_h\|_{H^1(\Omega)} \\ &\leq \frac{M}{\alpha} \|u - \Pi_h^r u\|_{H^1(\Omega)}\end{aligned}$$

Using (10):

$$\|u - u_h\| \leq C \frac{M}{\alpha} \left(\sum_{K \in \mathcal{T}_h} h_K^{2r} |u|_{H^{r+1}(\Omega)}^2 \right)^{1/2} \quad (12)$$

Using (11):

$$\|u - u_h\| \leq C \frac{M}{\alpha} h^r |u|_{H^{r+1}(\Omega)} \quad (13)$$

Error estimate for $\|u - u_h\|_{L^2(\Omega)}$ (not for exam)

Adjoint form

Consider a bilinear form $a: V \times V \rightarrow \mathbb{R}$. The *adjoint form* a^* is defined as:

$$a^*: V \times V \rightarrow \mathbb{R}$$

$$a^*(v, w) = a(w, v) \quad \forall v, w \in V$$

Remark: If a is symmetric, then $a^* = a$

$\forall g \in L^2(\Omega)$, consider the adjoint problem:

$$\begin{cases} \text{Find } \phi = \phi(g) \in V \\ a^*(\phi, v) = (g, v) = \int_{\Omega} g v \quad \forall v \in V \end{cases} \quad (14)$$

Assume that $\phi \in H^2(\Omega) \cap V$ (elliptic regularity)

Example

Consider $\mathcal{L} = -\Delta$. Then the solution of

$$\begin{cases} -\Delta \phi = g & \text{in } \Omega \\ \phi = 0 & \text{on } \partial\Omega \end{cases}$$

satisfies $\phi \in H^2(\Omega)$. Moreover:

$$\exists C_1 > 0: \quad \|\phi(g)\|_{H^2(\Omega)} \leq C_1 \|g\|_{L^2(\Omega)} \quad (15)$$

Take now $g = e_h = u - u_h$ in (14). Then:

$$\begin{aligned}\|e_h\|_{L^2(\Omega)}^2 &= (e_h, e_h) = a^*(\phi, e_h) = a(e_h, \phi) \\ &= a(e_h, \phi - \phi_h) \quad (\text{Galerkin orthogonality, for } \phi_h \in V_h) \\ &\leq M \|e_h\|_{H^1(\Omega)} \|\phi - \phi_h\|_{H^1(\Omega)}\end{aligned}$$

Take $\phi_h = \Pi_h^1 \phi$. Then:

$$\begin{aligned}\|e_h\|_{L^2(\Omega)}^2 &\leq M \|e_h\|_{H^1(\Omega)} \|\phi - \Pi_h^1 \phi\|_{H^1(\Omega)} \\ &\leq M \|e_h\|_{H^1(\Omega)} C_2 h |\phi|_{H^2(\Omega)} \quad (\text{for (11) with } m = r = 1) \\ &\leq M \|e_h\|_{H^1(\Omega)} C_2 h C_1 \|e_h\|_{L^2(\Omega)} \quad (\text{for (15)})\end{aligned}$$

Whence:

$$\begin{aligned}\|e_h\|_{L^2(\Omega)} &\leq M C_1 C_2 h \|e_h\|_{H^1(\Omega)} \\ &\leq M C_1 C_2 h C_3 h^r |u|_{H^{r+1}(\Omega)} \quad (\text{for (13)})\end{aligned}$$

Conclusion:

$$\|e_h\|_{L^2(\Omega)} \leq \overline{C} h^{r+1} |u|_{H^{r+1}(\Omega)} \quad (16)$$

A more general FE error estimate

Suppose $u \in H^{p+1}(\Omega)$, with $p \geq 0$.

Convergence rate for $\|u - u_h\|_V$:

	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p > 3$
$r = 1$	conv.	$\mathcal{O}(h)$	$\mathcal{O}(h)$	$\mathcal{O}(h)$	$\mathcal{O}(h)$
$r = 2$	conv.	$\mathcal{O}(h)$	$\mathcal{O}(h^2)$	$\mathcal{O}(h^2)$	$\mathcal{O}(h^2)$
$r = 3$	conv.	$\mathcal{O}(h)$	$\mathcal{O}(h^2)$	$\mathcal{O}(h^3)$	$\mathcal{O}(h^3)$

- Optimal rate of convergence for $p = r$.
- Suboptimal rate if $p < r$.
- Waste of computational effort if $p > r$ (extra regularity not rewarding in terms of order of accuracy).

Remark

For $\|u - u_h\|_{L^2}$, increase by one order the convergence rates of the table.